Tank 11: Use materials wisely

Goal: I can choose materials that do the job while reducing unnecessary waste.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 18, grade 6 path; Playbook B meeting 19, grade 6 path

Ask | Plan | Build | Test | **Improve**

You are done when

Show a useful material choice and explain what you saved or improved.

You will use

- Clean cardboard
- Paper
- Pencils
- Rulers
- Student scissors
- Masking tape
- Reusable trays
- Safe material samples
- Prototype
- Shared scale if available

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Compare two materials doing the same small job.
- Notice what can be reused or made smaller.

4 Step 4: Make it work — 17 min

- Sort the design's parts into essential and optional.
- Compare two suitable materials for one part.
- Improve that part without making the product harder to use.

5 Step 5: Check one thing — 5 min

- Check that the main job still works.
- Count pieces or compare material size before and after.
- Sort useful scraps for another group.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show a useful material choice and explain what you saved or improved.

What to turn in

One part made smaller or reused, noted in the folder.

No computer? Another way to do it

Compare materials by touch and weight only.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.